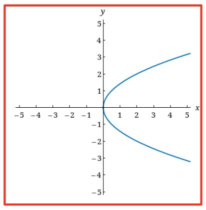
11.1

1) Match the equation with the graph

$y^2 = 2x$

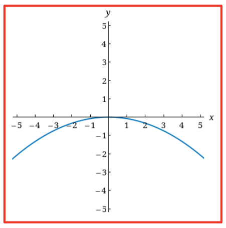


opens to the right

focal diameter: 2

2) Match the equation with the graph.

$12y + x^2 = 0$



opens down

focal diameter: 12

3) An equation of a parabola is given.

$x^2 = 12y$

Find focus, directrix, focal diameter

Focus: $4p = 12$

$p = 3$

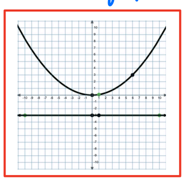
$\boxed{(0, 3)}$

Directrix: $y = -p$

$\boxed{y = -3}$

focal diameter: $|4p| = \boxed{12}$

Sketch a graph



4) An equation of a parabola is given.

$y = -4x^2$

Find focus, directrix, focal diameter

$x^2 = \frac{y}{-4}$

$x^2 = -\frac{1}{4} \cdot y$

Focus: $4p = -\frac{1}{p}$

$p = -\frac{1}{16}$

$\boxed{(0, -\frac{1}{16})}$

Directrix: $y = -p$

$y = -(-\frac{1}{16})$

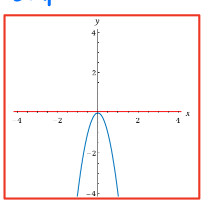
$\boxed{y = \frac{1}{16}}$

Focal diameter: $|4p|$

$= |-\frac{1}{4}|$

$= \boxed{\frac{1}{4}}$

Graph



5) An equation of a parabola is given.

$9x + 5y^2 = 0$

Find focus, directrix, focal diameter

$y^2 = -\frac{9}{5}x$

Focus: $4p = -\frac{9}{5}$

$p = -\frac{9}{20}$

$\boxed{\left(-\frac{9}{20}, 0\right)}$

Directrix: $x = -p$

$x = -\left(-\frac{9}{20}\right)$

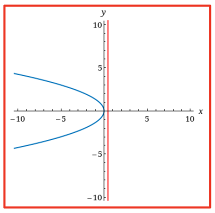
$\boxed{x = \frac{9}{20}}$

Focal diameter: $|4p|$

$= \left|-\frac{9}{5}\right|$

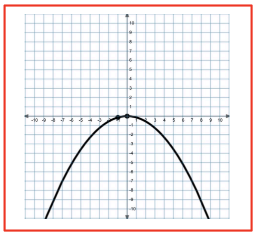
$= \boxed{\frac{9}{5}}$

Graph



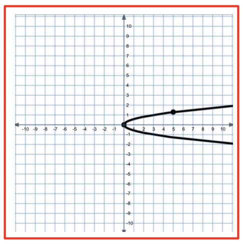
6) Use graphing Calculator to graph

$x^2 = -7y$



7) Use graphing calculator to graph

$x - 3y^2 = 0$



8) Find standard equation for parabola that has vertex at origin and satisfies given equation

Focus $F(0, 5)$

$p = 5$

$4p = 4(5) = 20$

$\boxed{x^2 = 20y}$

9) Find standard equation for parabola that has vertex at origin and satisfies given equation

Focus $F\left(-\frac{1}{28}, 0\right)$

$p = -\frac{1}{28}$

$4p = 4\left(-\frac{1}{28}\right) = -\frac{1}{7}$

$\boxed{y^2 = -\frac{1}{7}x}$

10) Find standard equation for parabola that has vertex at origin and satisfies given equation

Directrix $x = \frac{1}{12}$

$-p = \frac{1}{12}$

$p = -\frac{1}{12}$

$4p = 4\left(-\frac{1}{12}\right) = -\frac{1}{3}$

$\boxed{y^2 = -\frac{1}{3}x}$

11) Find standard equation for parabola that has vertex at origin and satisfies given equation

$y=-2$

$-p=-2$

$p=2$

$4p=4(2)=8$

$\boxed{x^2=8y}$

12) Find standard equation for parabola that has vertex at origin and satisfies given equation

Focus on positive x-axis, 2 units away from directrix

p is +, $y^2=4px$

$2|p|=2$

$|p|=1$

$p=1$

$4p=4$

$\boxed{y^2=4x}$

13) Find standard equation for parabola that has vertex at origin and satisfies given equation

Opens upward with focus 3 units away from vertex

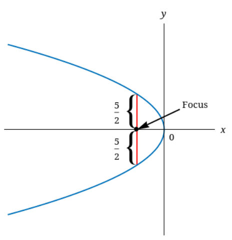
$x^2=4py$

p is +

$4p=4(3)=12$

$\boxed{x^2=12y}$

14) Find standard equation of parabola whose graph is shown.



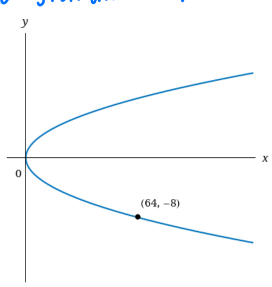
$|4p|=\frac{5}{2}\cdot 2$

$|4p|=5$

$4p=-5$

$\boxed{y^2=-5x}$

15) Find standard equation of parabola whose graph is shown.



$(-8)^2=4p(64)$

$4p=1$

$\boxed{y^2=x}$